Standard Operating Procedure for Resolving Major Optical Transport Failures on Cisco PE Equipment at Marathahalli Core (MAR-PE-13)

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_7964

Description

The failure domain of Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Cisco PE equipment in the Marathahalli Core site can be attributed to the inherent vulnerabilities in the optical transceivers and their susceptibility to environmental stressors. The Cisco PE equipment, being a critical component in the packet transport network, relies heavily on the stability of the optical signals received from the Optical Transport Network (OTN) equipment. Any degradation in the SFP/XFP or fiber cuts can lead to a cascade of errors in the packet forwarding process, resulting in the ALARM_7964 alarm being triggered. Furthermore, the OTN equipment's reliance on the Cisco PE equipment for packet routing and switching increases the likelihood of network-wide disruptions in the event of a major optical transport failure. Therefore, it is essential to promptly identify and resolve these failures to maintain network uptime and prevent data loss.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Marathahalli Core site for on-site assistance.

- Verify the status of the redundant link to ensure it is functioning correctly.

- Check the power supply and environmental conditions at the site to rule out any external factors contributing to the failure.

- Ensure the network management system (NMS) is properly configured to receive and display alarm information.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to high temperature and humidity levels.	High
Fiber cuts due to human error or environmental factors.	Medium
Rx/Tx power loss due to faulty OTN equipment or misconfigured Cisco PE settings.	Low

Diagnostic Commands & Expected Outputs

```
Command: show controller e1/0/0, Expected Output: Controller information for interface e1/0/0, Description: This command is used to gather information about the interface and its associated controller, which can help identify any issues with the SFP/XFP or fiber connectivity.
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```
Command: show run | include optical-power, Expected Output: Optical power settings for the interface, Description: This command is used to verify the optical power settings for the interface and check for any misconfigurations that may be contributing to the Rx/Tx power loss.

Command: show inventory | include SFP, Expected Output: SFP information, including serial number and status, Description: This command is used to gather information about the SFP/XFP, including its serial number and status, which can help identify any issues with the SFP/XFP or its connection to the Cisco PE equipment.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP with a new one, Command: no service old-mib, Description: This command is used to disable the old MIB to prevent any potential issues with the new SFP/XFP.

Step 2: Step: Verify SFP/XFP installation, Command: show inventory | include SFP, Description: This command is used to verify that the new SFP/XFP has been properly installed and is functioning correctly.

Step 3: Step: Configure optical power settings, Command: configure terminal ciscoasa(config)# optical-power e1/0/0 0, Description: This command is used to configure the optical power settings for the interface to ensure proper transmission and reception of optical signals.

Post-Check Verification

```
Command: show interface e1/0/0, Expected Output: Interface information for e1/0/0, including line protocol status, Description: This command is used to verify the line protocol status of the interface and ensure that it is functioning correctly.

Command: show run | include optical-power, Expected Output: Optical power settings for the interface, Description: This command is used to verify that the optical power settings have been properly configured and are functioning correctly.

Command: show inventory | include SFP, Expected Output: SFP information, including serial number and status, Description: This command is used to verify that the new SFP/XFP is functioning correctly and is properly connected to the Cisco PE equipment.
```

Roll-back Procedures

- Step: Restore original SFP/XFP, Command: no service old-mib, Description: This command is used to restore the original SFP/XFP and revert any changes made during the troubleshooting process.

- Step: Reconfigure optical power settings, Command: configure terminal ciscoasa(config)# no optical-power e1/0/0 0, Description: This command is used to reconfigure the optical power settings to their original values and prevent any potential issues with the Cisco PE equipment.

- Step: Re-enable old MIB, Command: service old-mib, Description: This command is used to re-enable the old MIB and prevent any potential issues with the Cisco PE equipment.